Data Model Documentation

Apex Manufacturing Executive Dashboard

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Prepared By: Hemanth Chelimala
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1. Purpose

The purpose of this document is to describe the logical data model implemented for the Apex Manufacturing Executive Dashboard.

A well-designed data model improves report performance, simplifies DAX calculations, supports accurate filtering, and provides a scalable foundation for business intelligence reporting.

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2. Data Modeling Approach

The dashboard follows a **Star Schema** data modeling approach.

The model consists of:

- Fact Tables (Business Transactions)
- Dimension Tables (Business Descriptions)

This structure minimizes redundancy, improves query performance, and follows Microsoft Power BI modeling best practices.

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3. Data Model Architecture

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                    Plants
                      │
                      │
                      │
Products ───────┐     │
                │     │
Machines ───────┼── Production
                │
Employees ──────┘

Products ──────────── Inventory
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Products ──────────── Quality

Suppliers ─ Purchase Orders ─ Products

Machines ────────── Maintenance
```

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4. Dimension Tables

Dimension tables provide descriptive business information used for filtering and grouping data.

Table	Business Purpose
Plants	Manufacturing plant information
Products	Product master data
Employees	Employee details
Machines	Machine information
Suppliers	Supplier master data

Characteristics:

- Low data volume
- Descriptive attributes
- Used for slicers and filters
- Connected to fact tables using primary keys

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5. Fact Tables

Fact tables contain measurable business transactions.

Table	Business Purpose
Production	Daily production records
Quality	Quality inspection records
Inventory	Inventory stock records
Purchase Orders	Procurement transactions
Maintenance	Machine maintenance activities

Characteristics:

- High transaction volume

- Numeric business measures
- Connected to dimensions using foreign keys
- Source of KPI calculations

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6. Relationship Design

Relationships were established between fact and dimension tables using primary and foreign keys.

From	To	Relationship
Plants	Production	One-to-Many
Products	Production	One-to-Many
Machines	Production	One-to-Many
Products	Inventory	One-to-Many
Products	Quality	One-to-Many
Suppliers	Purchase Orders	One-to-Many
Products	Purchase Orders	One-to-Many
Machines	Maintenance	One-to-Many

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7. Relationship Cardinality

The model primarily uses **One-to-Many (1:*)** relationships.

Advantages include:

- Efficient filtering
- Simplified DAX calculations
- Better report performance
- Reduced ambiguity

Many-to-Many relationships were avoided to maintain a clean and optimized model.

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8. Filter Direction

The model uses **Single Cross Filter Direction** wherever appropriate.

Benefits include:

- Faster query execution
- Reduced relationship ambiguity

- Improved model stability
- Easier maintenance

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9. Key Design Principles

The following modeling best practices were applied:

- Star Schema architecture
- Fact and Dimension separation
- Unique primary keys
- Foreign key relationships
- Business-friendly table names
- Consistent naming conventions
- Minimized data redundancy
- Optimized model for Power BI

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10. Data Flow

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Microsoft Excel Dataset

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Power Query (ETL)

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Data Cleaning

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Data Model

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Relationships

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DAX Measures

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Power BI Dashboard

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Business Insights
```

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11. Benefits of the Data Model

The implemented data model provides:

- Improved dashboard performance
- Faster report refresh
- Simplified maintenance
- Accurate KPI calculations
- Consistent filtering
- Better scalability
- Reliable business reporting

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12. Data Modeling Best Practices

The following Microsoft Power BI best practices were followed:

- Use of Star Schema
- Separate Fact and Dimension tables
- Descriptive table naming
- Single-direction filtering
- Unique primary keys
- Avoid unnecessary relationships
- Minimize calculated columns where possible
- Use measures for business calculations

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13. Future Enhancements

The data model can be extended by adding:

- Calendar (Date) Dimension
- Shift Dimension
- Customer Dimension
- Production Line Dimension
- Warehouse Dimension
- Cost Center Dimension
- Row-Level Security (RLS)
- Incremental Refresh
- Composite Models

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14. Conclusion

The Apex Manufacturing Executive Dashboard is built on a structured Star Schema data model designed for efficient business intelligence reporting. By separating transactional fact tables from descriptive dimension tables and implementing well-defined relationships, the model supports accurate analytics, scalable reporting, and optimized Power BI performance.

This data model provides a robust foundation for interactive dashboards and enables stakeholders to analyze production, quality, maintenance, inventory, and procurement data with confidence.